

MICHAEL LIU

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Education

University of Waterloo

Sept 2025 - Apr 2030

BCS in Computer Science, Digital Hardware Specialization

- Major Average: 93.1% | Cumulative Average: 92.7% | Student ID: 21177966
- Activities: Varsity Cross-Country, Varsity Track, WAT.ai

Job Experience

Machine Learning Research Assistant

Toronto, ON

The Hospital for Sick Children (SickKids)

May 2026 - Aug 2026

- Researching reaction-based GFlowNet models for molecular design and drug discovery.

Machine Learning Scientist

Waterloo, ON

WAT.ai

Sept 2025 - Feb 2026

- Collaborated with a small team of students on a medical imaging project focused on improving the segmentation of microaneurysms in fundus imaging.
- Co-authored a research paper detailing our methodology and results.

Tennis Instructor

Unionville, ON

Unionville Tennis Club

Apr 2022 - Aug 2025

- TPA Certified Instructor, 500+ Hours On-Court.

Projects

CourtVision: Computer Vision for Tennis Analytics

[github/tennis-cv-analysis](#)

OpenCV, PyTorch, YOLO, MediaPipe, XGBoost, Pandas, SciPy



- Built a computer vision system for tennis match analysis including player/ball tracking, court keypoint detection, and shot detection from broadcast camera footage.
- Implemented pose-based shot classification and computed player and ball speed statistics displayed through annotated match video output.

ArXtract: AI Search Engine for ML Research

[arxtract-cxc.vercel.app](#)

React, TypeScript, FastAPI

- Built a full-stack ML research tool using retrieval-augmented generation to query, rank, and summarize arXiv papers.
- Engineered semantic search using vector embeddings and cosine similarity to extract relevant sections from machine learning research papers based on user prompts.

HydraLA-Net Optimization for Microaneurysm Segmentation

[github/fundus-image-segmentation](#)

PyTorch, NumPy, Albumentations, Pandas, scikit-learn, Streamlit



- Built PyTorch training pipeline and trained HydraLA-Net (U-Net variation) segmentation models for retinal lesion detection.
- Conducted ablation studies on preprocessing and loss-function design to improve small lesion segmentation.

Technical Skills

Programming Languages: C/C++, Python, SQL, TypeScript

Data/ML: PyTorch, scikit-learn, Pandas, NumPy, Matplotlib, LLM/GenAI, OpenCV, YOLO

Databases/Tools: PostgreSQL, MySQL; Git, GitHub; LaTeX; Excel, Tableau

Frontend: React, HTML, CSS